



A review on 3D Gaussian splatting for sparse view reconstruction

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Abstract

Sparse view 3D reconstruction remains challenging due to inherent data scale limitations. Mainstream sparse view 3D reconstruction algorithms based on the NeRF framework struggle to balance generation quality and real-time performance. Recently, the advent of 3D Gaussian Splatting technology has demonstrated remarkable results, becoming increasingly prominent in 3D scene representation and reconstruction. Exploring the application of 3D Gaussian Splatting technology for sparse view 3D reconstruction represents a promising research avenue. Based on this, our paper provides a comprehensive review of current sparse view 3D reconstruction methods leveraging 3D Gaussian Splatting, with an emphasis on extracting effective reconstruction information from input images and utilizing these data to generate realistic scenes efficiently and reliably. We then provide a detailed discussion on how the algorithm addresses issues such as artifacts and scale ambiguous, which are common challenges in this field. In the subsequent sections, we present both quantitative and qualitative comparisons of various sparse-view 3D reconstruction methods, roughly demonstrating the advantages of sparse view 3D Gaussian splatting methods in terms of reconstruction quality and efficiency. Furthermore, we analyze the potential applications of sparse view 3D Gaussian splatting methods. Finally, we identify the challenges faced by sparse-view 3D Gaussian splatting reconstruction and suggest potential solutions. We hope that our analysis will provide valuable insights for future research efforts.

Keywords 3D gaussian splatting · Sparse view · Few shot · 3D reconstruction · Computer graphics

1 Introduction

3D reconstruction is the process of capturing the shape and appearance of real objects (Moons et al. 2010). 3D reconstruction algorithms generate 3D scenes from 2D images or video streams, with the quality of reconstruction heavily reliant on the comprehensiveness of the input data in covering the object features. In scenarios with limited input information,

such as restricted access, reconstruction quality deteriorates significantly. Sparse view 3D reconstruction algorithms are meticulously designed to reconstruct 3D scenes from one or a few images (Li et al. 2024), addressing the challenge of limited sample input. Nonetheless, due to the constraints on input data, these algorithms tend to overfitting during the reconstruction process (Zhu et al. 2023; Chung et al. 2024). Moreover, sparse view 3D reconstruction frequently encounters issues such as artifacts (Zhang et al. 2024; Li et al. 2024; Xu et al. 2024c), scale ambiguity (Zhu et al. 2023; Chung et al. 2024; Charatan et al. 2024), multi-view inconsistency (Feng et al. 2024; Shen et al. 2024) as the complexity of the reconstruction task and algorithmic limitations. Current research predominantly leverages advanced deep learning techniques to guide or directly construct sparse view 3D reconstruction algorithms, aiming to enhance algorithm efficiency while ensuring reconstruction accuracy (Fei et al. 2024; Chen and Wang 2024; Wu et al. 2024c; Bao et al. 2024; Dalal et al. 2024).

3D Gaussian Splatting (3D GS) (Kerbl et al. 2023) is an innovative method for 3D scene representation, incorporating differentiable rendering and adaptive algorithms, primarily used for efficiently and accurately modeling 3D scenes from static scenic images, facilitating the rapid accumulation of high quality 3D digital assets. 3D GS replaces traditional rasterized triangular meshes (Müller et al. 2022) or voxels (Fridovich-Keil et al. 2022) with anisotropic 3D Gaussian ellipsoids (referred to as 3D Gaussians, or Gaussians), optimizing their parameters through differentiable rendering methods differ from neural networks. Additionally, 3D GS employs adaptive algorithms to optimize the distribution of 3D Gaussians. By fully utilizing GPU rasterization pipeline technology, 3D GS significantly enhances algorithm efficiency (Lassner and Zollhofer 2021). Compared to implicit radiance field algorithms like Neural Radiance Fields (NeRF) (Mildenhall et al. 2021), 3D GS's explicit point-based radiance field (Chen and Wang 2024) representation substantially reduces overhead in low-density scenarios (e.g., air).

As a pioneering algorithm, the advantages of 3D GS in sparse view 3D reconstruction tasks remain unexplored. This paper focuses on the application of 3D GS in sparse view 3D reconstruction, providing a systematic review of existing research to facilitate a deeper understanding of 3D GS-based sparse 3D reconstruction algorithms and promote further investigation. The main contributions of this paper are as follows:

- We systematically review recent sparse view 3D reconstruction algorithms based on 3D GS, integrating them into a unified framework.
- We review sparse view 3D GS reconstruction algorithms from three perspectives: multi-dimensional feature extract, 3D Gaussian generation and anomaly handling.
- We perform a preliminary quantitative and qualitative analysis of existing sparse view 3D reconstruction algorithms based on known results.
- We explore the potential future applications of sparse-view 3D GS reconstruction methods, identify the challenges and difficulties faced by these algorithms, and provide valuable research directions and insights for future exploration.

The structure of this review is organized as follows: Sect. 2 introduces the backgrounds of sparse view 3D reconstruction and 3D GS, along with commonly used 3D datasets and loss functions; Sect. 3 deconstructs the sparse view 3D GS algorithms, outlining their typical constituent modules; Sect. 4 explores how sparse view 3D GS algorithms extract multi-

dimensional feature; Sect. 5 discusses various algorithms for generating 3D Gaussians in sparse view 3D reconstruction; Sect. 6 addresses algorithms and techniques for handling anomaly in sparse view 3D reconstruction; Sect. 7 compares the quantitative and qualitative experimental results of sparse view 3D GS algorithms with other approaches; Sect. 8 analyzes the future potential applications of sparse view 3D GS algorithms; the final section discusses the limitations of applying 3D GS in sparse view 3D reconstruction tasks and proposes potential solutions.

2 Backgrounds

2.1 Sparse view 3D reconstruction

Early approaches to sparse view 3D reconstruction include Structure-from-Motion (SfM) (Snavely et al. 2006), Multi-View Stereo (MVS) (Aliev et al. 2020; Kopanas et al. 2021; Rückert et al. 2022), real-world scanning (Arayici 2007), and Multi-Plane Images (MPI) (Zhou et al. 2018; Mildenhall et al. 2019; Wizarwongsa et al. 2021). SfM estimates camera positions and scene structure by extracting and matching key points across multiple views, while MVS further densifies these results to generate more detailed 3D models. Despite their effectiveness in capturing geometric information, these methods typically require a large number of input views and are computationally intensive, particularly in complex scenes. Real-world scanning provide robust 3D priors for neural networks but are generally limited to specific object categories. MPI represents the views by generating sub-images at different depths; however, while effective for depth and color re-rendering, it performs poorly in handling large disparities or when only a few views are available.

The introduction of NeRF (Mildenhall et al. 2021) marked a significant breakthrough in 3D reconstruction and novel view synthesis. NeRF leverages neural networks to represent the color and geometry of a scene, utilizing volumetric rendering techniques to produce high quality 3D scenes, particularly in densely sampled view scenarios. Nevertheless, NeRF has notable limitations, including the requirement for a large number of input views to achieve high quality reconstruction and slow rendering speeds, which hinder its practical utility. To adapt NeRF for sparse view scenarios, researchers have proposed various enhancements. Methods like DietNeRF (Jain et al. 2021) reduce inter-view variation by introducing constraint mechanisms, while approaches such as SparseNeRF (Wang et al. 2023) regularize outputs by incorporating depth priors. These methods enhance semantic consistency and geometric detail but may be limited in handling low-level detail and depth estimation quality.

To address NeRF's shortcomings, researchers have explored alternative representations, such as voxel grids (Fridovich-Keil et al. 2022), triplane representations (Zhou et al. 2018; Chan et al. 2022), and 3D GS (Kerbl et al. 2023). Voxel grids store color and opacity information in a regular 3D grid, offering fast rendering speeds suitable for low-resolution scenes, though they suffer from high memory demands and poor performance in complex scenes. Triplane representations decompose information into three orthogonal planes, providing high rendering quality with reduced memory usage, yet may be less effective in handling complex scenes or large disparity variations. 3D GS represents scene with Gaussians, enabling extremely fast rendering, particularly for real-time applications, but may produce floating artifacts and lose local detail when faces with sparse view.

Moreover, to enhance 3D reconstruction performance, researchers have proposed various optimization techniques, including incorporating depth information (Chung et al. 2024; Li et al. 2024; Zhu et al. 2023), using Transformer architectures (Zou et al. 2024; Xu et al. 2024b; Chen et al. 2024), regularization strategies (Li et al. 2024; Feng et al. 2024; Chen and Wang 2024) and diffusion models (Tang et al. 2023; Xiong et al. 2023; Feng et al. 2024). These carefully designed techniques are integrated into the sparse view 3D reconstruction process, maximizing their benefits for specific task scenarios. These techniques broaden the applicability of reconstruction algorithms and will be further discussed in the subsequent sections. Recent innovations focus on improving generalization (Lee et al. 2024; Yang et al. 2024a), reducing artifacts (Li et al. 2024; Xiong et al. 2023; Zhu et al. 2024a), and achieving real-time rendering (Zhang et al. 2024; Pan et al. 2024; Guo et al. 2024), making these methods more feasible for practical applications.

2.2 3D Gaussian splatting

3D GS (Kerbl et al. 2023) represents a radiance field explicitly by discretizing it as a collection of three-dimensional Gaussian distributions. Specifically, 3D GS approximates the color and opacity distributions of the radiance field using Gaussian ellipsoids, while performing 3D-to-2D rendering via view transformations and projections. The radiance field in 3D GS can be expressed as Eq. 1:

$$\mathcal{R}_{GS}(\mathbf{x}) = \sum_{i=1}^N \alpha_i \cdot \mathcal{G}_i(\mathbf{x}) \cdot c_i, \quad (1)$$

where $\mathbf{x} \in \mathbb{R}^3$ represents a 3D spatial coordinate in the scene, and $\mathcal{G}_i(\mathbf{x})$ is the distribution function of the i -th Gaussian ellipsoid at the point $\mathbf{x}$. The color c_i corresponds to the color of the i -th ellipsoid, $\alpha_i \in [0, 1]$ is its opacity, and N is the total number of Gaussian ellipsoids in the scene.

Each Gaussian ellipsoid is described by a three-dimensional anisotropic Gaussian distribution, defined as Eq. 2:

$$\mathcal{G}_i(\mathbf{x}) = \exp \left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_i)^\top \boldsymbol{\Sigma}_i^{-1} (\mathbf{x} - \boldsymbol{\mu}_i) \right), \quad (2)$$

where $\boldsymbol{\mu}_i \in \mathbb{R}^3$ denotes the center of the ellipsoid, and $\boldsymbol{\Sigma}_i$ is the covariance matrix, which as follows:

$$\boldsymbol{\Sigma}_i = \mathbf{R}_i \mathbf{S}_i \mathbf{S}_i^\top \mathbf{R}_i^\top, \quad (3)$$

where the covariance matrix is defined as the product of a scaling matrix $\mathbf{S}_i$, a rotation matrix $\mathbf{R}_i$, and their respective transposes, as illustrated in Eq. 3. The scaling and rotation matrices can be represented by a scaling factor and a quaternion, respectively. The differential forms of these parameters had been derived in the 3D Gaussian paper to facilitate their updating during the training process. This process of rotation and scaling endows the covariance matrix with physical interpretability while ensuring its positive semi-definite nature.

When projecting the 3D scene onto a 2D plane, both view transformation and projection are applied. To preserve the Gaussian characteristics of each Gaussian ellipsoid during the projection process, the affine approximation of the Jacobian matrix $J_i \in \mathbb{R}^{2 \times 3}$ is employed. The corresponding covariance matrix Σ'_i is then transformed as follows:

$$\Sigma'_i = J_i W_i \Sigma_i W_i^\top J_i^\top, \quad (4)$$

where $W_i \in \mathbb{R}^{3 \times 3}$ represents the view transformation matrix.

The algorithm encodes the RGB color values of the ellipsoids using three 4th-order spherical harmonics. The rendered color of each pixel is determined through iterative optimization to minimize rendering loss.

The rendering equation for the radiance field employs an α -blending algorithm, which accumulates the contributions of the discrete Gaussian ellipsoids to calculate the pixel value of any given pixel C , as shown in Eq. 5:

$$C = \sum_{i=1}^{N'} c_i \alpha'_i T_i, \quad (5)$$

and

$$\alpha'_i = \alpha_i \times \exp \left(-\frac{1}{2} (x' - \mu'_i)^\top (\Sigma'_i)^{-1} (x' - \mu'_i) \right), \quad (6)$$

$$T_i = \prod_{j=1}^{i-1} (1 - \alpha'_j), \quad (7)$$

where N' denotes the sorted Gaussian ellipsoid list in given pixel, (x', μ'_i, Σ'_i) are approximations derived based on a projection transformation, α'_i (Chen and Wang 2024) denotes the opacity after projection, T_i denotes the cumulative transparency, accounting for attenuation as the ray traverses through multiple ellipsoids.

The 3D GS pipeline involves several key stages, beginning with point cloud generation. Initial point cloud data is generated using the SfM (Snavely et al. 2006) algorithm based on input image sets. In this process, each point is substituted with a 3D Gaussian ellipsoid, which is assigned initial color and opacity values. The covariance matrix for these ellipsoids is initialized as an isotropic Gaussian, allowing for a fundamental representation of their spatial properties.

Following point cloud generation, the training phase commences. Here, 3D-to-2D rendering is performed using a defined rendering algorithm. The rendered 2D images are then compared with ground truth images to compute a loss function, facilitating the adjustment of Gaussian ellipsoid parameters through gradient descent. The set of ellipsoids is dynamically refined, incorporating new ellipsoids or removing existing ones based on the quality of the reconstruction.

To enhance computational efficiency, a tiling strategy is employed, which divides the screen into 16×16 pixel tiles. Ellipsoids that lie outside the view frustum are removed, and

the remaining ellipsoids within each tile are assigned IDs and sorted by depth using a fast GPU radix sort. For each pixel, contributions are computed from front to back, which are subsequently combined using the α -blending algorithm to achieve a coherent rendering.

To further improve reconstruction quality, ellipsoids are adaptively added or split in response to local gradients within the view space. In under-reconstructed regions, smaller ellipsoids are cloned and shifted along the gradient direction, while in over-reconstructed areas, larger ellipsoids are split into smaller ones, with appropriate scaling applied. This adaptive refinement process ensures an optimal distribution of Gaussians within the 3D space, ultimately enhancing overall reconstruction fidelity.

2.3 Datasets

In the context of sparse view 3D GS reconstruction, various datasets offer abundant support for both research and application. These datasets encompass a wide array of domains, ranging from multi-view images (Deitke et al. 2023) to dynamic human motion capture (Joo et al. 2015), facilitating a multitude of tasks such as novel view synthesis, 3D reconstruction, and instance segmentation. Table 1 presents detailed information about these datasets, including their scale, samples and basic description.

Panoptic Studio (Joo et al. 2015) is a dataset capturing social interactions with 480 synchronized cameras, providing 3D skeletal reconstructions and motion trajectories. It includes a variety of social game scenes, such as bargaining and prisoner's dilemma, and provides temporally synchronized 3D skeletal reconstructions, annotated motion paths, and camera calibration information.

CO3D (Reizenstein et al. 2021) is a large-scale dataset featuring 50 object categories with 19,000 video segments and 1.5 million frames. It provides multi-view 3D reconstruction and novel view generation, making it ideal for research in 3D scene modeling and view synthesis.

ZJU-MoCap (Peng et al. 2021) contains multi-view videos of 9 dynamic human performances, captured with 21 cameras. The dataset provides camera poses and human segmentation, making it essential for studying dynamic human neural representations and 3D reconstruction.

Google Scanned Objects (GSO) (Downs et al. 2022) includes over 1,000 3D models of household objects with textures and physical properties. It is useful for robotic navigation, grasping, and training in interactive 3D simulations.

Mip-NeRF 360 (Barron et al. 2022) focuses on 360-degree boundaryless scenes for view synthesis. Optimized for complex indoor and outdoor environments, Mip-NeRF 360 is suitable for 3D scene modeling and view generation.

Objaverse (Deitke et al. 2023) is a large dataset with over 800,000 3D models across 21,000 categories. It supports 3D generation, segmentation, and open-vocabulary navigation tasks, serving as a key resource for 3D vision research.

Objaverse-XL (Deitke et al. 2024) is an expansive dataset with 10 million 3D objects from platforms like GitHub and Thingiverse. It supports generative AI, 3D reconstruction, and novel view generation, advancing 3D vision models.

OmniObject3D (Wu et al. 2023) contains 6,000 high-quality real-world scanned 3D objects across 190 categories. This dataset offers data for 3D tasks like view synthesis, surface reconstruction, and 3D generation.

Table 1 3D reconstruction datasets used in sparse view tasks

Dataset	Scale	Samples	Description
Panoptic Studio (Joo et al. 2015)	480 Views	Multiple Categories	High resolution multi-view dataset for studying human activities
CO3D (Reizenstein et al. 2021)	40,000 Videos	50 Categories	Object-centric videos with accurate 3D reconstruction data
ZJU-MoCap (Peng et al. 2021)	9 Videos	21 Cameras	Use a multi-camera system to capture videos of complex dynamic humans
GSO (Downs et al. 2022)	1,000 Objects	17 Categories	Google high quality 3D scanned household items dataset
Mip-NeRF 360 (Barron et al. 2022)	362–1,107 Images	9 Categories	High quality image dataset for training neural radiance fields
Objaverse 1.0 (Deitke et al. 2023)	800K Objects	Multiple Categories	Contains over 800K 3D assets designed for a variety of AI research and applications
Objaverse-XL (Deitke et al. 2024)	10.2M Objects	800+ Categories	Internet 3D model library Contains highly diverse 3D data collected from multiple Internet sources
OmniObject3D (Wu et al. 2023)	6,000 Objects	190 Categories	Contains a large scale vocabulary of real world scans and annotations
Hi4D (Yin et al. 2023)	100 Sequences	20 pairs of Interacting Subjects	Contains high quality 4D scans of closely interacting humans
OpenIllumination (Liu et al. 2024a)	108K Images	64 Objects, 24 Different Materials	Contains high quality real object image datasets with multiple materials and lighting conditions
ShapeNet-SRN (Sitzmann et al. 2019)	350K Images	4500 Chairs, 2500 Cars	A high-quality collection of multi-category 3D models designed for multi-view consistency evaluation and geometric scene reconstruction
ACID (Liu et al. 2021)	13,000 Videos	765 aerial landscape videos	With a focus on 3D scene reconstruction and video synthesis tasks for coastlines and natural scenery
RealEstate10k (Zhou et al. 2018)	7,000 Videos	750,000 Frames	Multi-view dataset mined from YouTube real estate videos, used for view synthesis and scene extrapolation research
D-NeRF (Pumarola et al. 2021)	8 dynamic scenes, 100–200 views	Sparse images per scene	High-quality dataset of dynamic scenes with rigid and non-rigid motions, used for training and evaluating dynamic neural radiance fields
Hyper-NeRF (Park et al. 2021)	2 cameras, videos	30–60 s sequences	Captures topologically dynamic scenes like peeling bananas or facial expressions
NeRF-DS (Yan et al. 2023)	8 Scenes, 500 Frames Each	Various Specular Objects	Dynamic specular dataset for evaluating neural radiance fields. Includes moving and deforming objects in realistic environments
NeRF-LLFF (Mildenhall et al. 2019)	20–30 Input Images per Scene	24 Different Scenes	A real-world dataset for novel view synthesis using sparse handheld captures, blending layered scene representations for high-quality rendering
DTU (Jensen et al. 2014)	4,000 Images	80 Scenes, 49/64 Camera Views	Large-scale multi-view stereo dataset with diverse scenes and precise 3D scans
Neu3DV (Li et al. 2022)	18–21 Views	Multiple dynamic scenes	Provides high-resolution dynamic multi-view data, covering complex dynamic scenes and supporting high-quality 4D reconstruction

Hi4D (Yin et al. 2023) focuses on close-range human interactions, providing over 11,000 frames from 20 pairs of participants. With 4D scans and segmentation data, it is ideal for studying dynamic human interactions and shape reconstruction.

OpenIllumination (Liu et al. 2024a) is a dataset with 64 objects captured in over 108,000 images from 72 viewpoints under diverse lighting. It aids research in inverse rendering and novel view synthesis under various lighting conditions.

ShapeNet-SRN (Sitzmann et al. 2019) includes 3D models from the ShapeNet v2 library, with random viewpoint images for 50 models per scene. It is used for novel view synthesis and 3D reconstruction research.

ACID (Aerial Coastline Imagery Dataset) (Liu et al. 2021) features aerial drone videos of coastlines, providing over 2.1 million frames. This dataset is used for training long-distance view generation models and scene modeling in natural environments.

RealEstate10K (Zhou et al. 2018) consists of 7,000 short real estate video clips, offering multi-view images and camera poses. It is ideal for tasks like novel view synthesis and view extrapolation in diverse indoor and outdoor scenes.

D-NeRF (Pumarola et al. 2021) is designed for reconstructing and rendering dynamic scenes involving glossy objects, captured in real-world environments through monocular RGB video. Each scene contains changes in the motion or deformation of objects, providing samples for training and testing. The dataset provides rendered images from 100 to 200 viewpoints at a resolution of 800x800 pixels.

HyperNeRF (Park et al. 2021) focuses on topological changes in dynamic scenes, captured from monocular video. This dataset supports research on interpolation and novel view generation in environments with significant geometric changes.

NeRF-DS (Yan et al. 2023) dataset is specifically designed for reconstructing and rendering dynamic scenes involving glossy objects, captured in real-world environments through monocular RGB video. Each scene contains changes in the motion or deformation of objects, providing samples for training and testing.

NeRF-LLFF (Mildenhall et al. 2019) captures multi-view images of natural scenes with a handheld camera, providing high-quality novel view generation for virtual exploration and view interpolation using MPI representations.

DTU (Jensen et al. 2014) is a multi-view stereo dataset with 80 scenes captured by 49 or 64 cameras. It includes diverse objects like fabrics and sculptures, supporting 3D reconstruction and multi-view geometry tasks.

Neu3DV dataset (Li et al. 2022) consists of multi-view, time-synchronized dynamic videos captured by 21 GoPro cameras, covering complex scene variations and topological changes. The videos have a resolution of 2028×2704 pixels, with a frame rate of 30 FPS, and the dataset includes accurate camera calibration data.

2.4 Commonly used loss functions

Loss functions play a pivotal role in sparse view 3D GS reconstruction algorithms, serving as essential metrics to evaluate reconstruction accuracy, geometric consistency, and perceptual quality. They provide critical constraints for the optimization process, enabling enhanced detail recovery, improved geometric coherence, and realistic rendering. Well-designed loss functions further guide depth estimation, semantic alignment, and mask-based refinements effectively. This section will introduce the commonly used loss functions.

$\mathcal{L}_{RGB}$ (Pan et al. 2024) represents the $\mathcal{L}_1$ loss in RGB space, capturing pixel-wise differences between the reconstructed image and the ground truth. $\mathcal{L}_{SSIM}$ measures the Structural Similarity Index (SSIM) (Wang et al. 2004), which evaluates the perceptual and structural similarity between the reconstructed image and the reference.

$$\mathcal{L}_{RGB} = \|I - \hat{I}\|_1, \quad (8)$$

$$\mathcal{L}_{SSIM} = 1 - \text{SSIM}(I, \hat{I}), \quad (9)$$

where $\mathcal{L}_{RGB}$ calculate the pixel-wise difference between the generated image I and the reference image $\hat{I}$, $\text{SSIM}(I, \hat{I})$ is the Structural Similarity Index, which evaluates the structural similarity between the generated image I and the reference image $\hat{I}$. Its range is $[0, 1]$, where higher values indicate greater similarity.

Structural Dissimilarity (D-SSIM) (Wang et al. 2004) serves as a complementary metric to SSIM, designed to quantify the structural differences between two images, similar to SSIM:

$$\mathcal{L}_{D-SSIM} = \frac{1 - \text{SSIM}(I, \hat{I})}{2}. \quad (10)$$

3D GS (Kerbl et al. 2023) introduces a reconstruction loss $\mathcal{L}_{recon}$, which comprises $\mathcal{L}_1$ and $\mathcal{L}_{D-SSIM}$, with α as a hyperparameter, defined as follows:

$$\mathcal{L}_{recon} = (1 - \alpha)\mathcal{L}_1 + \alpha\mathcal{L}_{D-SSIM}, \quad (11)$$

Learned Perceptual Image Patch Similarity (LPIPS) (Johnson et al. 2016) measures visual similarity between two images, providing results closer to human perception. Unlike traditional metrics such as Peak Signal to Noise Ratio (PSNR) (Tanchenko 2014) and SSIM, LPIPS captures complex perceptual details using deep network features for more precise similarity measurement.

The LPIPS distance d between two images x and x' is computed as follows:

$$d(x, x') = \sum_l \frac{1}{H_l W_l} \sum_{h=1}^{H_l} \sum_{w=1}^{W_l} \|w_l \odot (\hat{y}_{l,hw} - \hat{y}'_{l,hw})\|_2^2, \quad (12)$$

where $\hat{y}_{l,hw}$ and $\hat{y}'_{l,hw}$ are the normalized feature activations at layer l and spatial location (h, w) , H_l and W_l are the height and width of the feature map at layer l , w_l is a learned weight vector applied to the features at layer l , $\odot$ denotes element-wise multiplication.

Score Distillation Sampling (SDS) (Poole et al. 2022) utilizes the score function in the pre-trained 2D diffusion model to optimize the parameters of the 3D reconstruction model, making the probability density of the generated data closer to that of the real data. Suppose θ are the parameters of the generative model, $\epsilon_\theta(z_t; t)$ is the noise predicted by the diffusion model at timestep t , SDS loss is expressed as:

$$L_{SDS}(\theta) = \mathbb{E}_{t, \epsilon} \left[w(t) (\epsilon_{\theta}(z_t; t) - \epsilon) \cdot \frac{\partial g(\theta)}{\partial \theta} \right], \quad (13)$$

where $g(\theta)$ is the output of the reconstruction model, $w(t)$ is a weighting function that controls the contribution of the loss at different time steps.

3 Analysis framework

To systematically review the existing sparse view 3D GS reconstruction algorithms, this paper consolidates these methodologies into a unified framework, summarizing them into four core steps: **Sparse inputs**, **Multi-dimensional Feature Extract**, **3D Gaussian Generation** and **Outputs**, as illustrated in Fig. 1, will be elaborated in Sect. 4, Sect. 5 and Sect. 6. Figure 2 showcases several representative sparse-view 3D GS reconstruction algorithms, each employing distinct strategies to optimize the 3D reconstruction process. It should be noted that not all algorithms strictly adhere to this framework and may differ in some specific details.

3.1 Sparse inputs

Sparse inputs forms the foundation of sparse view 3D reconstruction, which is characterized by its reliance on a limited number of input images. This distinguishes it from other reconstruction methods and typically encompasses three scenarios: single view (Szymanowicz et al. 2024), few-shot (Xiong et al. 2023), and monocular video (Zhang et al. 2024). Monocular videos are typically segmented into frames for integration into subsequent steps. The

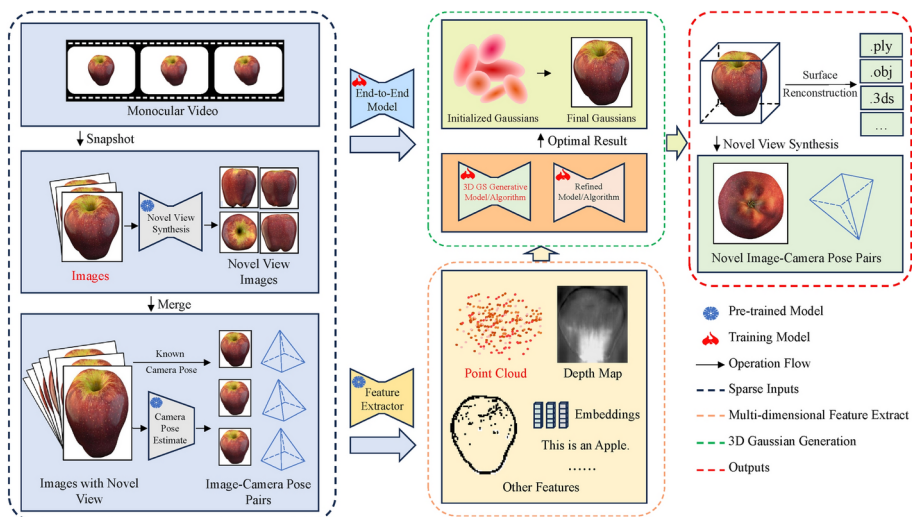


Fig. 1 The schematic diagram illustrates the framework of sparse view 3D reconstruction algorithms. The red text represents the modules corresponding to each step of the 3D GS algorithm, while the others are extensions introduced in subsequent research. Several illustrations of trapezoidal structures in various colors represent the encoder-decoder network. It is important to note that a sparse-view 3D GS reconstruction algorithm does not necessarily need to incorporate all of these elements

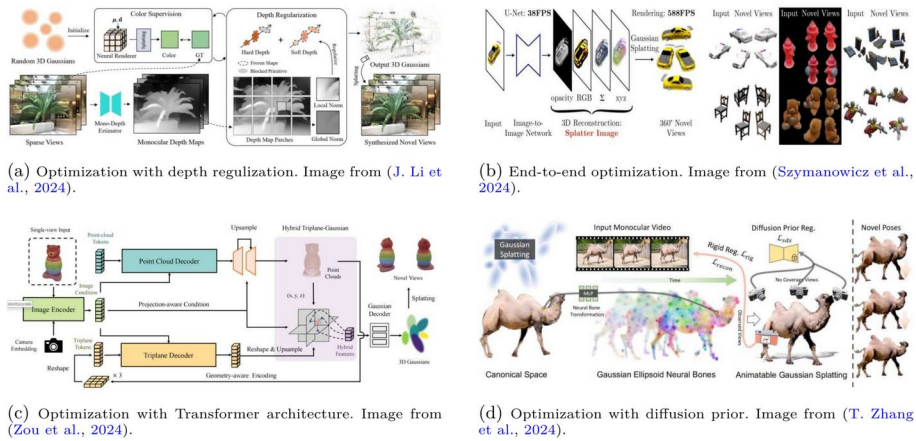


Fig. 2 Typical optimization strategies for sparse view 3D GS reconstruction

commonality among these scenarios lies in the absence of complete target view information within the input data, necessitating supplementation during the 3D modeling process. Currently, sparse view 3D reconstruction primarily focuses on single objects, with input data sourced from capturing the physical world or generated by novel view synthesis models (Fei et al. 2024). Camera pose estimation typically relies on additional camera pose estimation algorithms (Xu et al. 2024a). Novel view synthesis and camera pose estimation in the input stage are optional.

3.2 Multi-dimensional feature extract

The primary objective of this stage is to extract essential modeling information from sparse input data to guide subsequent 3D Gaussian generation. This phase depends on algorithms and models for point cloud generation, depth estimation, and feature extraction to obtain point clouds (Fisher et al. 2021), depth maps (Bhat et al. 2023), and multi-dimensional feature information (Oquab et al. 2023). Section 4 will discuss methods for acquiring these diverse dimensions of image information, while Sect. 5 will explore how to leverage them to guide the modeling of 3D Gaussians.

3.3 3d Gaussian generation

The captured modeling information is utilized as input for generative model to generate 3D Gaussians, with the distribution of these Gaussians in 3D space and their specific parameters being determined at this stage. Researchers have proposed various generative models (Yang et al. 2024b; Shen et al. 2024; Tang et al. 2023), typically aligning with the feature extractor outputs and depth maps referenced above. At times, the results generated by the generative model require further refinement and enhancement through the use of a refined model. Additionally, some algorithms are designed to generate 3D Gaussians entirely in an end-to-end (Szymanowicz et al. 2024; Wu et al. 2024a; Charatan et al. 2024) fashion from the input data. This step involves the generative model and 3D Gaussian generation, to be elaborated upon in Sect. 5. The refined models are employed to address anomalies such as

artifacts (Li et al. 2024) and scale ambiguity (Charatan et al. 2024) that occur during the previous reconstruction process, will be discussed in Sect. 6.

3.4 Outputs

The output stage typically utilizes surface reconstruction algorithms, such as the marching cubes (Lorensen et al. 1998; Yang et al. 2024c) algorithm, Poisson surface reconstruction (Kazhdan et al. 2006; Guédon and Lepetit 2024; Wu et al. 2024d), to extract 3D models from the 3D Gaussians, which can then be exported as 3D assets in various usable formats. These 3D models can also be leveraged to generate novel view images, akin to the functionality of NeRF.

4 Multi-dimensional feature extract

The 3D GS approach employs SfM to extract point clouds from input images, forming a skeletal transformation from 2D images to 3D space. With abundant view information, SfM can generate dense point clouds and high quality renderings. However, under sparse view conditions, SfM lacks the necessary input to construct comprehensive point clouds, resulting in sparse clouds that fail to meet the requirements of subsequent computations (Zhu et al. 2023). Additionally, sparse views pose significant challenges for accurately modeling complex object surfaces. Relying solely on sparse point clouds often does not achieve the level of detail required for fine modeling (Chung et al. 2024). Therefore, more robust and refined feature extraction is needed to capture usable features from the limited input images. Recent algorithms leverage feature extractors to process image data, providing extensive feature information to guide the reconstruction process, thereby significantly improving reconstruction quality. Table 2 summarizes the methods proposed in Chapter 4.

4.1 Point cloud

When reconstructing with sparse point clouds, the 3D GS algorithm tends to use large Gaussians to supplement invisible view domains, leading to defects in the geometry, texture, edges, and material of the reconstruction target. Therefore, dense point clouds are essential for effective 3D GS reconstruction. Zou et al. (Zou et al. 2024) initially generate a coarse point cloud from a single image using a point cloud decoder, followed by a two-step Snowflake Point Deconvolution (SPD) (Xiang et al. 2021, 2022) module to densify the point cloud, increasing the number of points from 2048 to 16384. During the upsampling process, the SPD module utilizes shape codes extracted from the coarse point cloud. These shape codes encompass global shape information and integrate local image features projected from the input image, ensuring the generated high resolution point cloud aligns closely with the input image, providing detailed geometric representation for reconstruction.

Endogaussian (Chen and Wang 2024) introduces a masked dense point cloud generation method, directly utilizing RGBD images to produce point clouds. This new approach bypasses the complex computations of feature matching and pose estimation, enhancing efficiency, especially in single view scenarios, ensuring dense point clouds under sparse

Table 2 Summary of multi-dimensional feature extraction in sparse view 3D reconstruction

Feature type	Definition/description	Methods/techniques	Challenges addressed
Point Cloud	Sparse 3D point representation derived from input images using SfM or other methods	Snowflake Point Deconvolution (Zou et al. 2024) for densification Masked dense point cloud generation (Chen and Wang 2024)	Overcoming sparse data coverage Improving reconstruction quality
Depth Map	Represents distance from the camera to each pixel in a scene, used for structural details and alignment	Depth priors and regularization (Chung et al. 2024) Gaussian-guided upsampling (Zhu et al. 2023) Planar scanning and cost volumes (Chen et al. 2024)	Relative depth scale ambiguities Sparse view depth estimation
Semantic Features	High-level abstract features derived from images to guide reconstruction	Pre-trained feature extractors (Oquab et al. 2023) Combination with camera parameter embeddings	Improving reliability in complex scenes
Spatial & Positional	Positional encodings and spatial relationships extracted to refine cross-view alignments	Learnable positional encoding (Xu et al. 2024c) Spatial embeddings combined with asymmetric U-Net (Tang et al. 2024)	Cross-view feature integration
Geometric Features	Structural priors capturing shape and contour details	Visual hulls for scaffolding (Yang et al. 2024b) Plane decomposition (Feng et al. 2024)	Handling occlusions Preserving geometric consistency
Spatio-temporal Features	Features derived from time-aware inputs (e.g., monocular video) to capture temporal dynamics	SyncDreamer-T model for temporal aggregation (Pan et al. 2024) Dynamic Gaussian updates (Pan et al. 2024)	Modeling dynamic scenes
Cross-view Features	Features extracted and matched across sparse view-points for consistency and improved detail	Multi-depth cost volume generation (Chen et al. 2024) Patch-based depth correlation loss (Xiong et al. 2023)	Addressing sparse input and ensuring multi-view consistency

views. By incorporating masks to handle occlusions, it reduces artifacts during generation and improves reliability.

4.2 Depth map

Depth maps, which represent the distance from the camera to each pixel in a scene, are crucial for providing detailed 3D structural information. Typically derived from monocular depth estimation algorithms, depth maps play a vital role in facilitating 3D reconstruction processes. They are also instrumental in enhancing multi-view feature extraction and matching, ensuring precise depth alignment and coherent reconstruction across different views.

4.2.1 Optimizing 3D structural information

The rich depth information contained in the depth map provides strong guidance for transforming target scenes from 2D images to 3D space, thereby enabling the 3D structure modeling of target scenes that better conforms to geometric and physical conditions. Chung et al. (Chung et al. 2024) utilize depth priors to refine 3D GS by incorporating depth loss and smoothness loss, alongside color and SSIM losses. This approach addresses the relative depth scale ambiguities inherent in monocular depth estimation by employing L1 loss

for depth alignment and applying smoothness loss to adjacent pixels in the depth map. Similarly, Xiong et al. (2023) propose estimating scale and displacement parameters to align depths in metric space, thereby avoiding the distortions that may arise from direct depth alignment. Furthermore, Zhu et al. (Zhu et al. 2023) leverages Dense Prediction Transformer (DPT) (Ranftl et al. 2021) for monocular depth estimation and employs a Gaussian-guided upsampling strategy to densify Gaussians, ensuring precise alignment of 3D and 2D depths using Pearson correlation.

4.2.2 Sparse view feature extraction and matching

Depth maps are also pivotal in enhancing sparse view feature extraction and matching by maximizing the information available from limited views and ensuring consistent depth alignment across different perspectives. In sparse view tasks, depth maps allow for the extraction of depth features from limited view information, enhancing the reconstruction process. Chen et al. introduce MVSpLat (Chen et al. 2024), which extracts depth features through planar scanning in cross-view matches and generates cost volumes for multi-depth cross-view feature similarity. This method uses a 2D U-Net for refinement, particularly improving depth estimation in textureless areas, leading to coherent depth estimation across views. Xiong et al. (Xiong et al. 2023) address the challenge of aligning monocular depth, which provides relative depth, with the anchored depth from COLMAP by incorporating patch-based depth correlation loss in their SparseGS method. This approach ensures that depth features are accurately aligned in sparse view tasks, correcting for potential depth errors and enhancing the overall reconstruction quality.

4.3 Other features

In addition to commonly used point clouds and depth maps, other types of features are also widely used. This section categorizes and explores various methods based on the types of features they extract, highlighting their contributions to the field of 3D reconstruction.

4.3.1 Semantic feature extract

Shen et al.'s Gamba (Shen et al. 2024) framework demonstrates the importance of semantic and spatial feature extraction in 3D reconstruction. DINO v2 (Oquab et al. 2023) is employed to extract these features from images, transforming them into tokens that are then combined with high dimensional camera parameter embeddings generated by an MLP. This combination enables the prediction of 3D Gaussian parameters, thereby enhancing the reliability and accuracy of the reconstruction.

Similarly, Zou et al. (2024) leverage pre-trained DINO v2 (Oquab et al. 2023) to generate patch-wise semantic feature tokens, which are further adjusted with camera characteristics to improve view perception. Charatan et al.'s pixelSplat (Charatan et al. 2024) approach also focuses on semantic feature extraction, utilizing pre-trained image encoders like ResNet-50 (He et al. 2016) or ViT-B/8 (Dosovitskiy 2020) to extract depth and semantic features, thus facilitating the conversion of 2D images into 3D representations.

4.3.2 Spatial and positional feature extract

Xu et al.'s GRM framework (Xu et al. 2024c) employs a convolutional image tokenizer and self-attention modules to facilitate cross-view feature exchange. A notable aspect of this approach is the use of learnable positional encoding, which effectively captures spatial relationships within the images, thereby improving the integration of cross-view features.

Tang et al.'s LGM (Tang et al. 2024) method combines an asymmetric UNet (Ronneberger et al. 2015) with DINO v2 (Oquab et al. 2023) to extract spatial features and camera pose embeddings. This approach enhances the stability and precision of 3D reconstruction by focusing on the spatial relationships inherent in the data.

4.3.3 Geometric feature extract

FDGaussian (Feng et al. 2024) introduces a method that decomposes sparse view 3D reconstruction into new view generation and subsequent reconstruction. It employs self-attention and cross-attention mechanisms to extract geometry-sensitive plane features, which, when combined with semantic conditions from input images, guide a pre-trained UNet (Ronneberger et al. 2015) in generating target view images.

Yang et al.'s Gaussian Object (Yang et al. 2024b) approach utilizes the Visual Hull derived from sparse view images as a geometric scaffold, providing structural priors that are crucial for initial sampling in 3D reconstruction. This method emphasizes the importance of geometric structure information in the reconstruction process.

4.3.4 Spatio-temporal feature extract

Efficient4D (Pan et al. 2024) focuses on reconstructing 3D objects from monocular video by improving the SyncDreamer (Liu et al. 2023a) algorithm with SyncDreamer-T. This approach aggregates spatiotemporal information to enhance the subsequent diffusion model generation, thereby improving the 3D reconstruction of objects across video frames.

4.3.5 Cross-view feature extract

Chen et al.'s MVSplat (Chen et al. 2024) method integrates CNNs and Transformers to facilitate cross-view feature extraction and exchange. By combining the strengths of convolutional and attention-based mechanisms, this approach significantly improves the extraction and integration of features from multiple views, which is critical for accurate 3D reconstruction.

5 3D Gaussian generation

3D GS is built upon the explicit 3D representation of 3D Gaussians, which relies on an efficient 3D Gaussian generation algorithm. Recent studies have primarily focused on directly or indirectly generating 3D Gaussians, enhancing the realism and real-time performance of the existing 3D GS algorithms.

5.1 End-to-end generation of 3D Gaussians

3D Gaussians are defined by a range of parameters, including the centers of Gaussian ellipsoids, opacity, covariance matrices, and spherical harmonics. These parameters collectively capture intricate physical surfaces of the real world, encompassing details such as texture, material properties, and depth information. End-to-end 3D Gaussian generation utilizes an implicit inference neural network to directly predict 3D targets from input images, substantially reducing modeling complexity.

Shen et al. (2024) conceptualized the densification of 3D Gaussians as a long sequence processing task. They integrated the long sequence generation framework Mamba (Gu and Dao 2023) with 3D GS, introducing Gamba for end-to-end single view 3D reconstruction. The core component of Gamba, termed Gamba Former, comprises multiple Gamba blocks, each containing a Mamba block, a linear projection module for feature processing, and pre-set operation blocks. Gamba predicts 3D Gaussian parameters by integrating input image labels, camera embeddings, and a set of learnable 3D Gaussian embeddings.

Szymanowicz et al. (2024) developed Splatter Image, which utilizes a U-Net network to directly predict 3D Gaussian parameters by reversing the rendering process. The U-Net generates a 3D Gaussian parameter map pixel-by-pixel from the input image, termed as the Splatter Image. The network then predicts depth and offset values to determine the spatial positions of the 3D Gaussians. Training occurs on 3D datasets with reconstruction loss calculated between input and generated images, supplemented by standard regularization techniques to manage extreme Gaussian sizes on the CO3D dataset (Reizenstein et al. 2021).

Charatan et al. introduced pixelSplat (Charatan et al. 2024), an end-to-end approach for predicting 3D Gaussians from images. The method starts with a pretrained encoder that encodes the input image into feature vectors. A polar attention mechanism is then used to address scale ambiguity by inferring the scale factor for each scene. A fully connected neural network predicts 3D Gaussians from the encoded features, with their positions determined by sampling the depth distribution along rays to avoid local minima.

Tang et al. proposed LGM (Tang et al. 2024), generating 3D targets from four orthogonal azimuth and fixed elevation images. An asymmetric U-Net extracts and fuses features from the images to predict 3D Gaussians. Plücker ray embeddings (Jia 2020; Sitzmann et al. 2021) encode camera poses, integrating pixel RGB values into a 9-channel feature map used as input for the U-Net. The network follows a similar architecture to prior work but incorporates self-attention in deeper layers to optimize memory usage. Features from the four input images are flattened and interconnected to propagate information across views, with the asymmetric U-Net handling high-resolution inputs while constraining the number of output Gaussians.

Wu et al. developed the DIG3D framework (Wu et al. 2024a), based on an encoder-decoder architecture. The encoder utilizes U-Net and DINOv2 to extract features and depth information from single view images, integrating them into a unified representation. The decoder employs a deformation transformer-based module to align 3D input points with image features under the fused single view conditions, iteratively refining the reconstruction of 3D Gaussians from random initializations on a layer-by-layer basis.

5.2 3D Gaussian generation with generative models

Point clouds, depth maps, and feature vectors derived from input images and camera poses are used by generative models to convert into 3D Gaussians, further refining the details. These generative models are typically based on neural networks such as diffusion models, transformers, and U-Nets.

5.2.1 Diffusion models

Sparse view 3D reconstruction utilizes pretrained diffusion models, initially designed for 2D image generation, to guide and enhance the reconstruction process by refining details. In sparse view scenarios, newly generated views from the diffusion model are incorporated into the training, enhancing overall performance.

Tang et al. introduced DreamGaussian (Tang et al. 2023), which integrates 3D GS with diffusion models, alongside efficient mesh extraction and UV space texture repair algorithms. The framework begins by generating a rough 3D Gaussian from text or images using a single-step SDS loss, optimized through random view rendering. Tang et al. found that multi-step SDS training did not enhance the clarity and detail of the Gaussian distribution, likely due to inconsistent 3D guidance across optimization steps. Consequently, mesh extraction and texture refinement are employed. Mesh extraction involves querying local densities after partitioning 3D space, followed by back-projection of RGB values onto the mesh surface. UV space texture repair adds noise to rough images rendered from UV maps and employs diffusion models to denoise these images. To mitigate artifacts introduced by SDS loss, pixel-wise MSE loss between rough and input images is utilized.

Efficient4D (Pan et al. 2024) reconstructs 3D objects from monocular video, incorporating an image generation and a reconstruction module. The image generation component synthesizes a sequence of images from varying temporal and spatial perspectives, with SyncDreamer-T aggregating spatiotemporal information from video data to condition the diffusion model, thereby guiding multi-view image generation. The reconstruction module extends 3D GS to 4D GS by incorporating time variables, using the generated images to facilitate temporal modeling.

Building Animatable 3D Gaussian Splatting (BAGS) (Zhang et al. 2024) reconstructs animated 3D models from monocular video by predicting invisible viewpoint images using a trained diffusion model. To improve prediction accuracy and maintain consistency, rigid regularization constraints (Luiten et al. 2024) derived from physical simulations ensure physical plausibility during transformations, minimizing artifacts.

Lee et al. (Lee et al. 2024) developed a method for reconstructing multi-person 3D scenes from monocular video. They leveraged the diffusion model's SDS loss for reconstructing occluded regions and employed text inversion to enhance the SDS process, effectively constraining the scope of SDS generation and avoiding arbitrary appearance artifacts.

5.2.2 Other models

Zou et al. (Zou et al. 2024) proposed a hybrid 3D GS primitive representation called the TriplaneGaussian, which utilizes transformer-based point cloud and triplane decoders to reconstruct rough point clouds and implicit triplanes from ViT-encoded image features.

Upsampling modules and geometric-aware encoding further refine these representations, integrating the optimized point clouds and implicit triplanes with a Gaussian decoder to complete the GS process, thereby achieving rapid and high-quality single view reconstruction. Local image features extracted by the Transformer facilitate projection-aware adjustments to the rough point cloud and implicit triplane, enhancing alignment and consistency with the input views.

Yang et al. introduced the Gaussian Object (Yang et al. 2024b), employing a Gaussian repair model to generate compensatory images that address defects in target reconstructions. The Gaussian repair model, comprising ControlNet (Zhang et al. 2023) and stable diffusion (Rombach et al. 2022), processes rendered images with added noise from 3D GS, outputs predictions, and computes fine-tuning losses against input images for training. The repair model rectifies missing image data through distance-sensitive sampling, aligning the target center point with an elliptical path defined by training views. Viewpoints outside the initial input coverage are selected to render repaired images.

Chen et al. proposed MVSplat (Chen et al. 2024), which extracts depth features through plane scanning and cross-matching feature information from multiple views at varying depth planes. A U-Net equipped with cross-view perception is then used to predict depth estimation maps and 3D Gaussian parameters, with the center positions determined by back-projecting the estimated depth maps.

FDGaussian (Feng et al. 2024) introduces a two-stage framework for modeling 3D targets from single images. Initially, it employs an orthogonal plane decomposition mechanism based on self-attention and cross-attention to extract 3D geometric features from 2D inputs, enabling the generation of consistent multi-view images from a single view point using a U-Net. In the subsequent stage, it enhances 3D GS reconstruction through polar attention, effectively integrating individual view images to improve reconstruction fidelity. FDGaussian significantly enhances single view reconstruction accuracy by preserving geometric consistency across multiple views.

6 Anomaly handling

Anomaly handling is a crucial step in sparse view 3D reconstruction, as it plays a significant role in mitigating unreasonable artifacts, such as artifacts or floaters, that may emerge during the process. Sparse view approaches, often dependent on monocular depth estimation algorithms, frequently encounter challenges in accurately determining object positions, resulting in scale ambiguity issues. Figure 3 illustrates the artifacts and scale ambiguity anomalies observed during the reconstruction process (Charatan et al. 2024). Current algorithms employ various techniques to partially alleviate these anomalies, enhancing the overall robustness of the reconstruction.

6.1 Artifacts

In 3D GS, artifacts such as floaters and inaccurate depth representations can substantially degrade the quality of the generated models. This section discusses various techniques employed to address these artifacts, enhancing the visual fidelity and realism of 3D recon-

Fig. 3 Schematic diagram of artifact and scale ambiguity. Images reference from pixelSplat (Charatan et al. 2024)

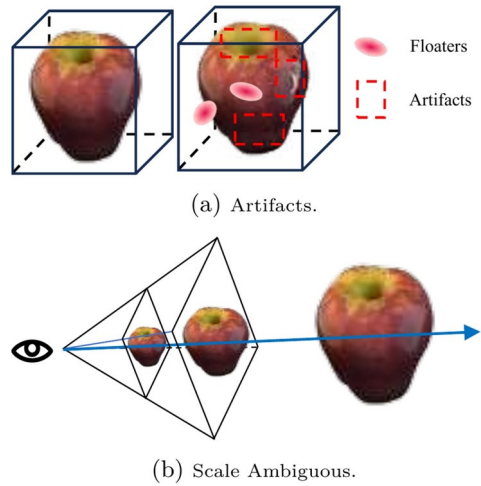
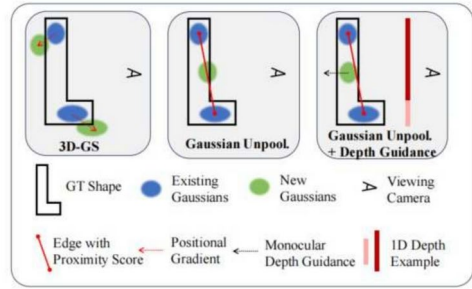
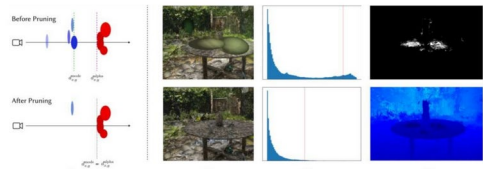


Table 3 Optimization and Solution of Artifact Problems by Different Methods, from (Xiong et al. 2023), (Feng et al. 2024), (Li et al. 2024), (Charatan et al. 2024), (Yang et al. 2024b), (Zhang et al. 2024), (Lee et al. 2024), (Tang et al. 2023), (Zhu et al. 2024a)

How to address artifact	Method	Description
Gaussian distribution	SparseGS	Deep guidance for pruning
	FDGaussian	Using GDS to minimize unnecessary segmentation and cloning operations
	DNGaussian	Deep regularization
	pixelSplat	Sampling from Probability Models
	GSOBJECT	Based on the obtained visual hull, exclude floaters according to the set threshold
View constraint	BAGS	Surface alignment, regularization
	pixelSplat, FDGaussian	Polar attention mechanism to effectively correlate features from different perspectives
	Lee et.al	Supplementary View of Diffusion Model
Piral Training strategy	GSOBJECT	Gaussian repair model
	Chung et.al	Early stop
	Dream Gaussian	The MSE loss function in the texture optimization stage is more suitable
	EndoGS	Total Variation Loss

structions. Table 3 provides a comprehensive summary of the relevant methods and classifies them based on their underlying approaches, while Fig. 4 illustrates two specific examples.

To eliminate floating artifacts in 3D Gaussian point clouds, the SparseGS approach (Xiong et al. 2023) calculates depth discrepancies using alpha blending and mode selection methods, followed by a dip test-based bivariate detection to evaluate the bivariate score of depth differences. Based on this score, an adaptive pruning threshold is established, creating

Fig. 4 Examples of anomaly handling**(a)** Eliminate artifact with depth guidance. Image from (Z. Zhu et al., 2023).**(b)** Floater Pruning. Image from (Xiong et al., 2023).

a mask to remove inaccurate Gaussian points. This effectively eliminates floating artifacts and enhances rendering quality.

Lee et al. (Lee et al. 2024) utilize a pretrained 2D diffusion model to generate images from novel viewpoints, calculating SDS loss to guide the optimization of 3D GS. This approach mitigates artifacts arising from sparse and occluded viewpoints, ensuring the consistency and realism of unobserved regions in the reconstructed 3D models.

FDGaussian (Feng et al. 2024) addresses artifacts while preserving geometric details by improving geometric consistency across images through geometry-aware multi-view generation. It introduces an epipolar attention mechanism to effectively correlate features from different viewpoints and employs Gaussian Divergence Saliency (GDS) to minimize unnecessary splits and cloning operations.

Chung et al. (Chung et al. 2024) employ an early stopping strategy to prevent overfitting by terminating the optimization process when depth-guided loss begins to increase. This approach helps halt optimization promptly when geometric guidance is ineffective, thus avoiding local optima and artifacts. The study also suggests that even rough point clouds can be instrumental in artifact elimination.

In DNGaussian (Li et al. 2024), hard depth regularization adjusts spatial positions by forcing nearby Gaussian bodies, potentially introducing artifacts or noise by overemphasizing Gaussians near surfaces. In contrast, soft depth regularization fine-tunes depth information by modulating the opacity of Gaussian bodies, effectively eliminating artifacts and ensuring rendering completeness and accuracy.

The pixelSplat method (Charatan et al. 2024) introduces a multi-view epipolar transformer to reliably infer scale factors for each scene, addressing scale ambiguities caused by arbitrary camera pose reconstruction. Additionally, implicit Gaussian sampling optimizes Gaussian parameters by avoiding local minima through sampling and reparameterizing the

probability distribution of 3D Gaussian positions, effectively reducing ghosting and motion blur artifacts.

DreamGaussian (Tang et al. 2023; Zhang et al. 2024) identifies that optimizing with SDS loss in the latent space during the initial stage can lead to overly saturated block artifacts on UV maps due to unstable gradients introduced during UV space texture optimization. Incorporating pixel-level MSE loss offers more stable and detailed gradients, resulting in smoother and more natural UV texture optimization, thereby enhancing texture detail and overall model quality.

In Yang et al.'s Gaussian Object (Yang et al. 2024b), the removal of floating artifacts utilizes a visual hull and K-nearest neighbors (KNN) algorithm. Initially, a visual hull is constructed using view cones and object masks from input images to approximate the geometric shape for initializing the 3D Gaussian representation. The KNN algorithm then calculates the average distance of each Gaussian point to its nearest neighbors, setting a statistically derived threshold to iteratively exclude floating Gaussians. Furthermore, Gaussian Object employs a Gaussian repair model to correct artifacts by generating degraded images through a leave-one-out strategy that introduces noise to Gaussian attributes. The pretrained Control Net model is fine-tuned with these images, enabling the repair model to convert degraded renderings into high-fidelity images, thus correcting artifacts in the Gaussian point distribution and enhancing rendering quality.

EndoGS (Zhu et al. 2024a) employs total variation loss in the spatiotemporal dimension to mitigate artifacts. By optimizing pixel values in occluded regions, total variation loss (Wu et al. 2024b) maintains color consistency and reduces color drift and artifacts caused by occlusions. This loss function smooths color transitions between adjacent pixels by minimizing image gradients, thereby eliminating high-frequency noise and unnatural boundaries, ultimately improving overall image quality.

BAGS (Zhang et al. 2024) incorporates surface alignment regularization loss to reduce artifacts by aligning 3D Gaussians with the reconstructed surface. Rigid regularization constraints ensure the physical plausibility of the reconstructed surface by preserving structural rigidity during transformations, further minimizing artifacts.

6.2 Scale ambiguity

Scale ambiguity is a persistent challenge in sparse view 3D reconstruction, where insufficient input data can lead to inaccuracies in size and distance perception within the model. This section examines strategies to address scale ambiguity, including depth regularization, epipolar geometry, and cross-view feature extraction, which collectively contribute to more consistent and accurate scale representations in reconstructed 3D models.

Chung et al. (2024) tackle scale ambiguity by generating dense depth maps using a pre-trained monocular depth estimation model and refining scale and offset with sparse depth points obtained from COLMAP. The approach minimizes discrepancies between sparse and dense depths by weighting sparse point reprojection errors, thereby mitigating scale ambiguity in monocular depth estimation.

SparseGS (Xiong et al. 2023) addresses scale ambiguity through the introduction of depth correlation loss and generative constraints. The depth correlation loss employs Pearson correlation coefficients to align alpha-blended depth maps with pretrained depth maps, thereby avoiding scale and displacement inconsistencies associated with direct MSE appli-

cation. Generative constraints further guide the generation of 3D Gaussian point clouds in underrepresented regions, reducing scale ambiguity and enhancing depth accuracy in new viewpoints.

PixelSplat (Charatan et al. 2024) resolves scale ambiguity using a multi-view epipolar transformer that reliably infers scale factors for each scene, ensuring consistent depth and scale representations in 3D reconstructions. Implicit Gaussian sampling predicts the probability distribution of Gaussian positions, facilitating accurate scale inference and optimization through differentiable sampling. This approach addresses scale ambiguity resulting from arbitrary camera pose reconstructions, thereby enhancing the precision and effectiveness of 3D reconstructions.

7 Performance comparison

This section presents a quantitative and qualitative comparison of sparse view algorithms, providing a detailed analysis of the results. Specifically, this includes a comparison of common metrics for these algorithms across various datasets and different sparse view configurations. Finally, a summary of the sparse view 3D GS methods discussed in the article is provided. The experimental results are sourced from the original papers, with all data appropriately cited. However, the reliability of these results is questionable due to inconsistencies in the hardware and software configurations, as well as model parameters mentioned in different studies. Therefore, these findings should be regarded as a reference for researchers.

We have created five tables to compare various 3D GS methods and relevant techniques. However, there are some limitations to this analysis. The effectiveness of these methods can vary due to factors such as environmental conditions, technical implementation, and user needs, which are difficult to quantify. Furthermore, rapid advancements in technology may mean some new methods are not included in this analysis. Therefore, while we aim for thoroughness in our comparison, readers should interpret the results with caution, considering the variability in real-world applications.

Table 4 compares the performance of NeRF-based methods and 3D GS-based methods on the MipNeRF360 dataset under 4-views conditions, using PSNR, SSIM, and LPIPS as evaluation metrics. The results indicate that models incorporating 3D GS demonstrate remarkable performance. Notably, the GSObject method achieved a PSNR value of 24.81, significantly surpassing that of the other methods, which underscores its superior image reconstruction capabilities.

Table 4 Quantitative results on MipNeRF360 (Barron et al. 2022) (4-views), in terms of PSNR, SSIM (Wang et al. 2004), and LPIPS (Zhang et al. 2018). LPIPS* equals to $LPIPS \times 100$. The methods are from (Sun et al. 2022), (Jain et al. 2021), (Niemeyer et al. 2022), (Yang et al. 2023), (Wang et al. 2023), (Shi et al. 2024), (Kerbl et al. 2023), (Zhu et al. 2023), (Yang et al. 2024)

Method	3D GS	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
DVGO		14.39	0.791	24.43
DietNeRF		18.90	0.897	11.17
RegNeRF		13.59	0.848	20.44
FreeNeRF		13.71	0.853	16.83
SparseNeRF		12.83	0.845	17.76
ZeroRF		14.17	0.819	19.88
3D GS	✓	20.31	0.899	10.80
FSGS	✓	21.07	0.910	9.51
GSObject	✓	24.81	0.935	4.98

Table 5 Quantitative results on NeRF-LLFF (Mildenhall et al. 2019) (3-views), in terms of PSNR, SSIM, LPIPS, training time, and FPS. The methods are from (Barron et al. 2022), (Jain et al. 2021), (Niemeyer et al. 2022), (Yang et al. 2023), (Wang et al. 2023), (Kerbl et al. 2023), (Chung et al. 2024), (Li et al. 2024)

Method	3D GS	PSNR↑	SSIM↑	LPIPS↓	Time↓	FPS↑
MipNeRF		14.62	0.351	0.495	2.2h	0.21
DietNeRF		14.94	0.370	0.496	–	0.14
RegNeRF		19.08	0.587	0.336	4 h	0.21
FreeNeRF		19.63	0.612	0.308	2.3h	0.21
SparseNeRF		<i>19.86</i>	<i>0.624</i>	0.328	1.5h	0.21
3D GS	✓	14.51	0.398	0.406	<i>2.7m</i>	280
Chung et al	✓	17.17	0.497	0.337	–	–
DNGaussian	✓	19.12	0.591	<i>0.294</i>	3.5m	<i>300</i>

Table 6 Quantitative results on GSO (Downs et al. 2022)(single view), in terms of PSNR, SSIM, LPIPS and inference time. The methods are from (Liu et al. 2023b), (Melas-Kyriazi et al. 2023), (Ye et al. 2024), (Long et al. 2024), (Liu et al. 2024b), (Jun and Nichol 2023), (Tang et al. 2023), (Zou et al. 2024), (Tang et al. 2024), (Xu et al. 2024c), (Feng et al. 2024)

Method	3D GS	PSNR↑	SSIM↑	LPIPS↓	Time↓
Zero-1-to-3		18.37	0.877	0.212	30 m
Realfusion		15.26	0.722	0.283	20 m
Consistent-123		19.46	0.858	0.146	–
Wonder3D		17.29	0.815	0.240	3 m
One-2-3-4-5		17.84	0.800	0.199	45 s
Shap-e		15.45	0.772	0.297	9 s
DreamGaussian	✓	19.56	0.853	0.174	2 m
TriplaneGaussian	✓	16.81	0.797	0.257	<i>0.2s</i>
LGM	✓	16.90	0.819	0.235	5 s
GRM	✓	20.10	0.826	<i>0.136</i>	5 s
FDGaussian	✓	22.98	<i>0.899</i>	0.146	70 s

Table 5 compares the performance of NeRF-based methods and 3D GS-based methods on the NeRF-LLFF dataset under 3-views conditions, using PSNR, SSIM, LPIPS, inference time and FPS as evaluation metrics. While RegNeRF, FreeNeRF, and SparseNeRF demonstrate strong capabilities across various performance metrics, their relatively low inference times and FPS render them unsuitable for real-time applications. In contrast, DNGaussian exhibits comparable performance to these traditional models but excels in real-time capabilities, achieving a commendable prediction speed.

Table 6 compares the performance of 3D GS-based methods and other approaches on the GSO dataset under single view conditions, using PSNR, SSIM, LPIPS, and inference time as evaluation metrics. FDGaussian demonstrates superior performance, achieving higher PSNR and SSIM values while maintaining competitive LPIPS performance. Additionally, TriplaneGaussian excels in real-time capabilities, managing to meet adequate thresholds across other metrics. Overall, the results indicate that methods based on Gaussian Splatting outperform their counterparts in both performance and efficiency.

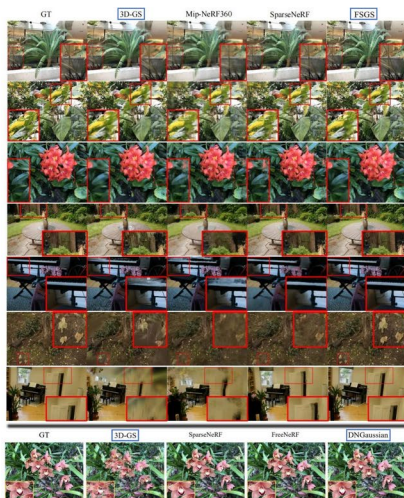
Table 7 compares the performance of 3D GS-based methods and other approaches on the D-NeRF dataset under monocular conditions, evaluating their effectiveness using PSNR, SSIM, FPS, and inference time as key metrics. While GS-based methods may not excel in performance metrics, they demonstrate a significant advantage in real-time capability. Notably, work of Katsumata et al. exhibit competitive reconstruction performance alongside commendable real-time efficiency.

Table 7 Quantitative results on D-NeRF (Pumarola et al. 2021) (monocular), in terms of PSNR, SSIM, FPS and training time. The methods are from (Fang et al. 2022), (Fridovich-Keil et al. 2023), (Gan et al. 2023), (Kerbl et al. 2023), (Luiten et al. 2024), (Katsumata et al. 2023)

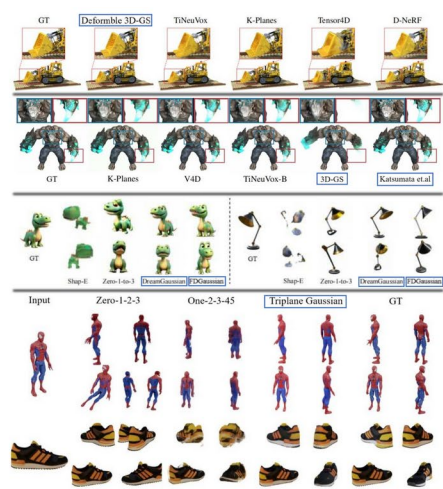
Method	3D GS	PSNR $\uparrow$	SSIM $\uparrow$	FPS $\uparrow$	Time $\downarrow$
TiNeuVox-S		30.75	0.96	0.32	8 m
TiNeuVox-B		32.67	0.97	0.13	28 m
K-Planes		31.61	0.97	0.54	52 m
V4D		33.72	0.98	1.47	6.9h
3D GS	✓	20.51	0.89	170	6 m
D-3D GS	✓	17.22	0.81	173	15 m
Katsumata et.al	✓	32.07	0.96	150	8 m

Table 8 Quantitative results on RealEstate10K (Zhou et al. 2018) (2-views(pairs)), in terms of PSNR, SSIM, LPIPS and inference time. The methods are from (Yu et al. 2021), (Suhail et al. 2022), (Du et al. 2023), (Xu et al. 2024b), (Charatan et al. 2024), (Chen et al. 2024)

Method	3D GS	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Time $\downarrow$
PixelNeRF		20.43	0.589	0.550	5.299s
GPNR		24.11	0.793	0.255	13.34s
AttnRend		24.78	0.820	0.213	1.325s
muRF		26.10	0.858	0.143	0.186s
pixelSplat	✓	25.89	0.858	0.142	0.104s
MVSplat	✓	26.39	0.869	0.128	0.044s



(a) Sparse view scene reconstruction. Images from (Z. Zhu et al., 2023), (J. Li et al., 2024).



(b) Sparse view target reconstruction. Images from (Z. Yang et al., 2024), (Katsumata et al., 2023), (Feng et al., 2024), (Zou et al., 2024).

Fig. 5 Comparison of sparse view 3D reconstruction. Methods based on 3D Gaussian Splatting are highlighted with a blue box. The red boxes highlight the regions of interest, showcasing key areas for detailed observation

Table 8 compares the performance of 3D GS-based methods and other approaches on the RealEstate10K dataset under 2-view conditions, using PSNR, SSIM, LPIPS, and inference time as key evaluation metrics. Two GS-based methods demonstrate outstanding reconstruction quality and real-time performance. Notably, muRF achieves comparable metrics.

Images in Fig. 5 are collected from the paper, covering both scene and object modeling. The corresponding algorithms are primarily based on 3D GS and NeRF frameworks.

Improved methods built upon 3D GS demonstrate superior realism, with significantly fewer artifacts, reduced floaters, and sharper edge contours. They also exhibit enhanced detail and texture fidelity, as exemplified by DNGaussian in (a) and FDGaussian and Triplane Gaussian in (b).

Table 9 summarizes the key methods discussed in the paper, including the modules employed (depth estimation, feature extraction, text prompts, diffusion models), loss functions, applied view scenes, and distinctive modules.

8 Application analysis

8.1 Robotic vision

In the realm of robotic vision, sparse view 3D GS has become an essential tool for real-time navigation and obstacle detection, thanks to its efficient 3D environmental reconstruction. By constructing simplified 3D models, robots can respond swiftly in complex and uncertain environments, significantly enhancing their autonomous decision-making and path-planning efficiency. In dynamic settings, sparse view 3D GS reduces computational resource consumption while providing highly accurate environmental models. However, a key challenge remains: in high-speed movements or particularly complex scenarios, sparse data may cause a decline in model precision, particularly in situations involving occlusion or substantial data loss, thus compromising the robot's decision-making capabilities. Future research must focus on improving the stability and robustness of sparse data processing, especially in dynamic environments, ensuring model accuracy and response speed (Zhu et al. 2024b).

8.2 Virtual reality

In the field of Virtual Reality (VR), sparse view 3D GS enhances the immersive user experience by efficiently rendering sparse data. As VR technology evolves and expands into increasingly complex scenes, the demand for computational power rises rapidly. Sparse view 3D GS enables precise rendering of large-scale environments, effectively avoiding delays and stuttering caused by hardware bottlenecks, thus enhancing the fluidity of the virtual environment. However, this technology still exhibits significant shortcomings when rendering large-scale dynamic scenes, particularly in complex interactions or detailed processing, where it often falls short of achieving ideal realism. Despite progress in existing research, the challenge remains to efficiently render large, dynamic environments while maintaining sufficient detail and realism. Future research should prioritize optimizing the rendering efficiency of expansive scenes while enhancing interactivity and sensory realism, striving to create virtual worlds that more closely resemble the physical world (Jiang et al. 2024).

8.3 Autonomous driving

In autonomous driving, sparse view 3D reconstruction can create real-time 3D models of the traffic environment, providing vital information about roads, pedestrians, and obstacles. This capability is crucial for decision-making in complex traffic situations. However, the

Table 9 Summary of sparse view 3D reconstruction methods leveraging 3D GS

Method	With depth estimation	With feature extract	With text prompt	With diffusion model	Loss functions	View number	Distinctive module
Chung et al. (Chung et al. 2024)	✓	✓			$\mathcal{L}_{RGB}, \mathcal{L}_{SSIM}, \mathcal{L}_{Depth}, \mathcal{L}_{Smooth}$	Few-shot	Deep guidance optimization strategy, early stop strategy and smoothness constraint
DNGaussian (Li et al. 2024)	✓	✓			$\mathcal{L}_{RGB}, \mathcal{L}_{Depth}, \mathcal{L}_2$	Few-shot	Hard/soft depth regularization, global/local depth normalization and neural color renderer
Deformable Gaussian (Yang et al. 2024a)		✓			$\mathcal{L}_1, \mathcal{L}_{D-SSIM}$	Few-shot	Deformable Gaussian pipeline and annealing smoothing training
FSGS (Zhu et al. 2023)	✓	✓			$\mathcal{L}_1, \mathcal{L}_{SSIM}, \mathcal{L}_{Depth}$	Few-shot	Proximity-guided gaussian unpooling

Table 9 (continued)

Method	With depth estimation	With feature extract	With text prompt	With diffusion model	Loss functions	View number	Distinctive module
SparseGS (Xiong et al. 2023)	✓	✓		✓	$\mathcal{L}_{RGB}, \mathcal{L}_{Depth}, \mathcal{L}_{SDS}$	Few-shot	Patch-based Depth Correlation Loss and Floater Elimination
Efficient4D (Pan et al. 2024)		✓		✓	$\mathcal{L}_1, \mathcal{L}_{SSIM}$	Monocular	4D Gaussian distribution representation and time-synchronous spatial volume
GSubject (Yang et al. 2024b)	✓	✓	✓	✓	$\mathcal{L}_{RGB}, \mathcal{L}_{Mask}, \mathcal{L}_{Depth}, \mathcal{L}_{LPIPS}$	Few-shot	Visual hull, gaussian repair model and "leave one out" training strategy
Lee et al. (Lee et al. 2024)	✓	✓	✓	✓	$\mathcal{L}_{MSE}, \mathcal{L}_{SSIM}, \mathcal{L}_{LPIPS}, \mathcal{L}_{SDS}, \mathcal{L}_{Depth}$	Few-shot	Canonical space optimization method

Table 9 (continued)

Method	With depth estimation	With feature extract	With text prompt	With diffusion model	Loss functions	View number	Distinctive module
DreamGaussian (Tang et al. 2023)		✓		✓	$\mathcal{L}_{SDS}, \mathcal{L}_2$	Single-view	Efficient mesh extraction algorithm from 3D Gaussians and a UV-space texture refinement
FDGaussian (Feng et al. 2024)		✓	✓	✓	$\mathcal{L}_{Recon}, \mathcal{L}_{SSIM}, \mathcal{L}_{LPIPS}$	Single-view	Orthogonal plane decomposition mechanism and two-stage framework epipolar attention mechanism
SplatterImage (Szymanowicz et al. 2024)	✓	✓			$\mathcal{L}_2, \mathcal{L}_{LPIPS}$	Single-view	Splatter Image
BAGS (Zhang et al. 2024)	✓	✓		✓	$\mathcal{L}_{SDS}, \mathcal{L}_{Rigid}, \mathcal{L}_1, \mathcal{L}_{Mask}, \mathcal{L}_{LPIPS}$	Monocular	Diffusion model and rigid regularization
Tri-planeGaussian (Zou et al. 2024)		✓			$\mathcal{L}_{CD}, \mathcal{L}_{EMD}, \mathcal{L}_2, \mathcal{L}_{Mask}, \mathcal{L}_{SSIM}, \mathcal{L}_{LPIPS}$	Single-view	Hybrid triplane gaussian

Table 9 (continued)

Method	With depth estimation	With feature extract	With text prompt	With diffusion model	Loss functions	View number	Distinctive module
GRM (Xu et al. 2024c)	✓	✓		✓	$\mathcal{L}_2, \mathcal{L}_{LPIPS}, \mathcal{L}_{Mask}$	Single-view	Pure transformer 3D GS architecture and pixel aligned gaussian method
pixelSplat (Charatan et al. 2024)	✓	✓			$\mathcal{L}_2, \mathcal{L}_{LPIPS}, \mathcal{L}_{Smooth}$	Few-shot	Epi-polar transformer and probabilistic primitive prediction
MVSplat (Chen et al. 2024)	✓	✓			$\mathcal{L}_2, \mathcal{L}_{LPIPS}$	Few-shot	Cost Volume
DIG3D (Wu et al. 2024b)	✓	✓			$\mathcal{L}_{RGB}, \mathcal{L}_{LPIPS}$	Single-view	Multi-level optimization of 3D Gaussian parameters and 3D reference point projection
Gamba (Shen et al. 2024)		✓			$\mathcal{L}_2, \mathcal{L}_{Mask}, \mathcal{L}_{LPIPS}$	Single-view	GambaFormer, mamba block designed for 3D GS

Table 9 (continued)

Method	With depth estimation	With feature extract	With text prompt	With diffusion model	Loss functions	View number	Distinctive module
LGM (Tang et al. 2024)	✓	✓	✓	✓	$\mathcal{L}_{RGB}, \mathcal{L}_2, \mathcal{L}_{LPIPS}$	Few-shot	Asymmetric U-Net attention propagation

technology faces challenges in integrating data from multiple sensors, especially when lighting and weather conditions change, which can reduce the stability and accuracy of the 3D models. While there has been progress in modeling urban streets and highways, performance remains inadequate in extreme weather (like heavy rain, fog, or nighttime) and complex traffic scenarios. Future research should focus on improving the adaptability and accuracy of sparse view 3D reconstruction in these challenging environments to ensure reliable environmental perception and effective decision-making in dynamic and adverse conditions (Chen and Wang 2024).

8.4 Medical imaging

In the field of medical imaging, sparse view 3D GS enables the reconstruction of fine biological tissue structures from limited imaging data, allowing doctors to make more accurate diagnoses. The efficient handling of sparse data ensures that high-quality reconstructions can still be achieved even under resource-constrained conditions, significantly improving diagnostic accuracy. However, challenges remain in the integration of multi-modal data (such as CT, MRI, ultrasound, etc.) in medical imaging. Current technologies struggle to accurately reconstruct high-resolution images from sparse or incomplete imaging data, particularly in complex biological tissue structures. Additionally, the real-time fusion of medical imaging and multi-modal data remains a significant hurdle. Future research should focus on enhancing the reconstruction accuracy of sparse data and improving the efficiency of multi-modal imaging processing to increase the comprehensiveness and real-time capabilities of clinical diagnoses (Zhu et al. 2024a; Zhou et al. 2023; Chen and Wang 2024).

9 Discussions and future works

This paper addresses sparse view 3D GS reconstruction, reviewing background knowledge related to 3D GS and sparse view 3D reconstruction. The study breaks down the sparse view 3D GS reconstruction task into four key areas: Sparse Inputs, Multi-dimensional Feature Extract, 3D Gaussian Generation, and Outputs. The paper discusses Multi-dimensional Feature Extraction, 3D Gaussian generation, and anomaly handling in detail. Finally, the paper identifies challenges within the field and outlines potential solutions.

9.1 Challenge 1: Limitations of Gaussian representations

Discrete Gaussian representations, while simplifying the modeling process, compromise the continuity and smoothness, leading to suboptimal performance in dynamic scenes. This limitation stems from the challenge of capturing the temporal evolution of Gaussian distributions, including their creation and dissipation over time.

9.2 Potential solutions

Introducing survival time parameters (start and end times) into the Gaussian representations could more accurately reflect the lifecycle of these distributions, enabling the model to dynamically manage topological changes, such as the emergence or removal of Gaussian distributions. Integrating Gaussian representations with NeRF models could further enhance rendering quality by leveraging NeRF's learned representations while maintaining the fast rendering capabilities of Gaussian techniques.

9.3 Challenge 2: Dependency on model performance. Dependency on model performance

The current approach heavily depends on auxiliary models, such as pose estimation, point cloud estimation, and diffusion models, with their performance directly impacting the final output quality. Regularization techniques provide limited mitigation.

9.4 Potential solutions

Enhancing the accuracy of pose and point cloud estimations, through optimizations in COLMAP could improve results, particularly in texture-scarce or complex surfaces. Furthermore, advanced geometric regularization techniques combined with multi-view inputs could bolster the model's robustness and generalization.

9.5 Challenge 3: Challenges in modeling special scenes

The model often struggles with complex scenes such as monochromatic spaces, specular reflections, and sky regions, where it fails to generate high-quality, realistic images. Addressing these scenarios requires the development of specific geometric priors or constraints.

9.6 Potential solutions

For example, incorporating priors like normal information or edge detection could better constrain Gaussian representations in challenging regions. In areas with specular reflections, integrating depth supervision or applying local depth normalization could mitigate surface inconsistencies caused by reflective effects.

9.7 Challenge 4: Resource consumption issues

The computational demands of handling complex scenes can significantly hinder practical applicability due to high memory and processing requirements.

9.8 Potential solutions

Distributed computing across multiple GPUs or CPUs can alleviate some of this load, improving processing efficiency while mitigating memory bottlenecks, albeit at the cost of increased computation time. Optimizing data representation and storage structures to minimize redundancy can further reduce resource consumption.

9.9 Challenge 5: Insufficient diversity in 3D datasets

The limited diversity of existing 3D datasets constrains the model's ability to generalize to real-world scenes.

9.10 Potential solutions

Future efforts should focus on developing larger and more varied 3D datasets, potentially by combining multiple scene-level datasets to enhance the training data's diversity. Additionally, data augmentation strategies that simulate a wider range of real-world variations could further strengthen the model's generalization capabilities.

9.11 Challenge 6: Overfitting in sparse views

Sparse view scenarios, especially those with limited viewpoint coverage, make models highly susceptible to overfitting, severely affecting generalization.

9.12 Potential solutions

Developing new geometric regularization methods that effectively exploit viewpoint information could reduce overfitting risks and enhance generalization performance.

9.13 Challenge 7: Challenges in complex texture generation

Despite advancements, current models continue to struggle with accurately generating complex textures, often resulting in outputs that lack realism.

9.14 Potential solutions

Enhancing Gaussian representation methods to improve density and detail capture could address these limitations. Incorporating high-resolution multi-view inputs could further increase sensitivity to fine details. Additionally, the application of deep learning-based image enhancement techniques during texture generation could significantly boost the realism of the rendered images.

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